

Debian Minimal GNOME Install: A Step-by-Step Tutorial

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Ubuntu-Style Minimal GNOME on Debian: How-To Guide

Published: Updated: 15201 views 4 mins read

If you're looking to install a lightweight version of GNOME on Debian, similar to Ubuntu's "Minimal Install," you're in the right place. This guide will walk you through the steps to perform a **Debian Minimal GNOME install** using Netinst ISO.

At the end, you will get a barebones GNOME environment up and running on your Debian system. Let's get started!

Step 1: Download the Netinst ISO

First, you'll need to download the Debian Netinst ISO. This is a minimal installation image that allows you to install Debian with just the basics.

Go to the [Debian download page](#) and grab the Netinst ISO file.

Next create a bootable USB drive with the ISO file using a tool like **Rufus** (for Windows) or [Ventoy](#) or [Etcher](#) (for Linux/Mac).

There are also so many other tools exist to help you create bootable USBs in Linux. If you're interested to know, check the following guides:

Command line bootable USB creation tools:

- [How To Create Bootable USB Drive Using dd Command](#)
- [Bootiso Lets You Safely Create Bootable USB Drive In Linux](#)

Graphical bootable USB creation tools:

- [Popsicle – Create Multiple Bootable USB Drives At Once](#)
- [Create Bootable USB Drive With USBImager In Linux](#)
- [Kindd – A Graphical Frontend To dd Command](#)

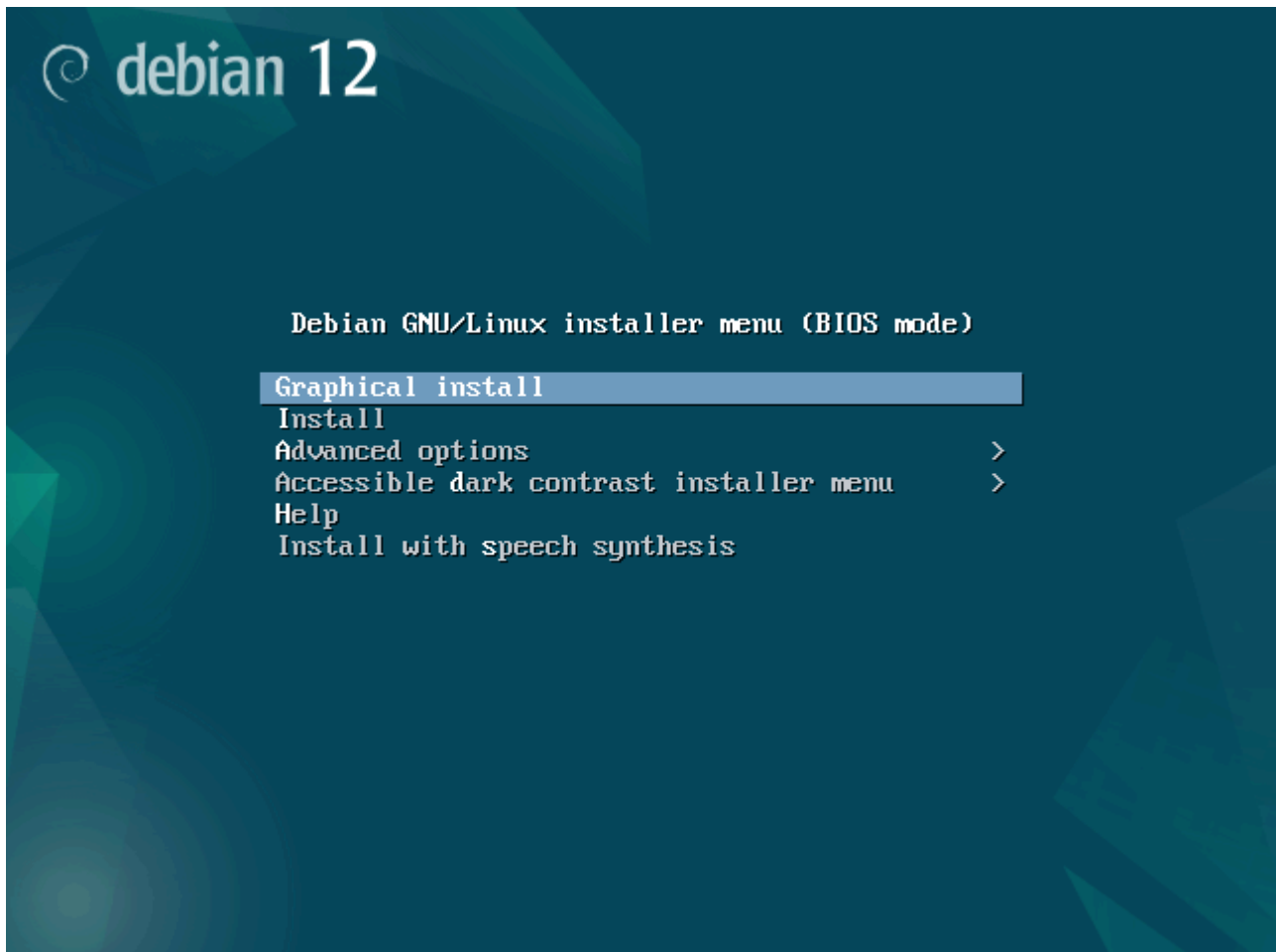
Step 2: Install Debian Minimal

1. Boot from the USB:

Insert the bootable USB into your computer and restart. Boot from the USB drive by selecting it in the BIOS/UEFI settings.

2. Start the Installation:

Start the [Debian installation](#) as usual and follow the on-screen instructions.



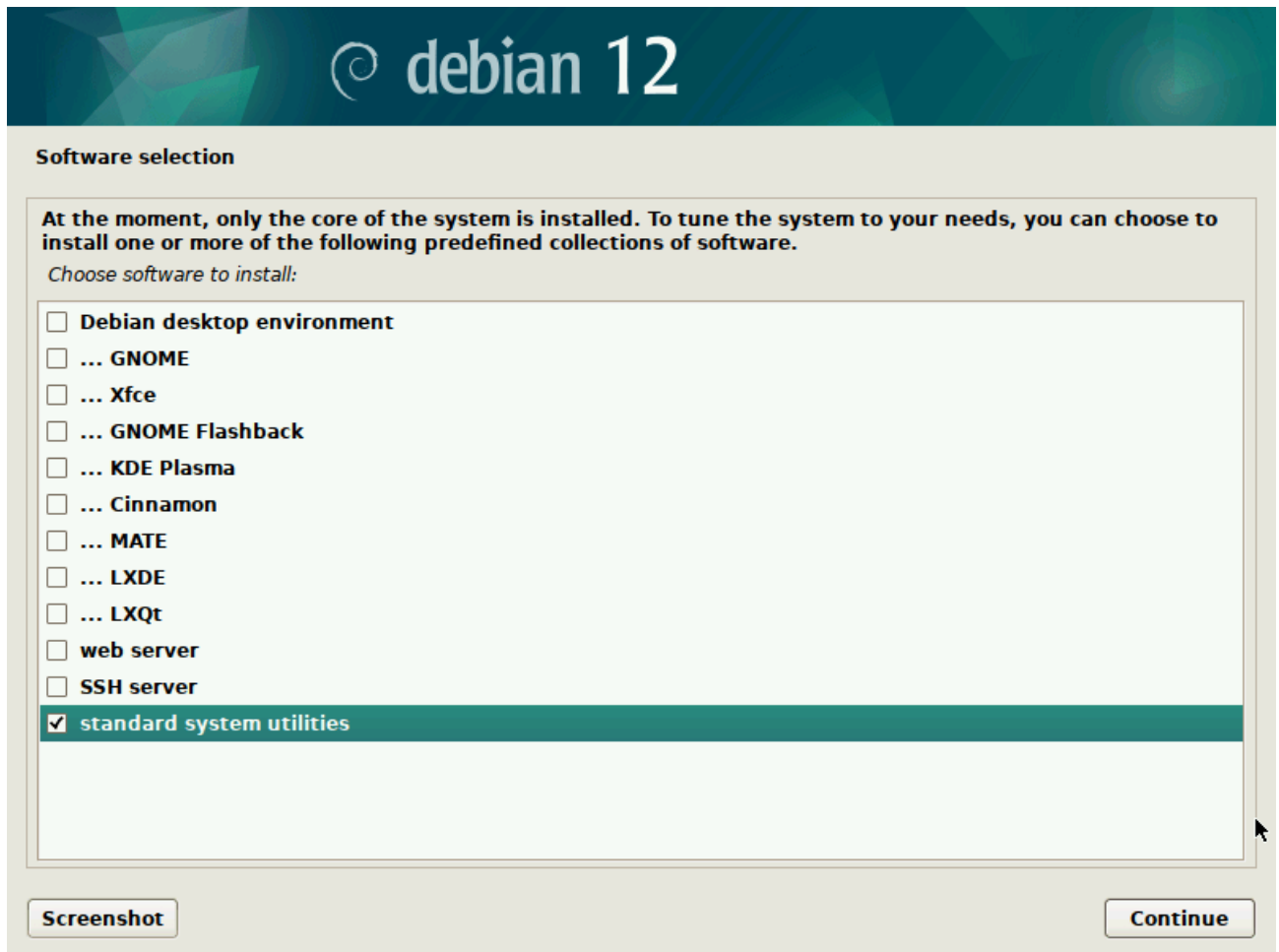
Start Debian Installation

Continue the installation process until you reach the software selection step.

3. Software Selection:

This is the IMPORTANT step.

When you get to the "Software Selection" step, **deselect everything** except "**Standard System Utilities.**"



Choose Standard System Utilities

This will ensure you only install the bare minimum.

4. Complete the Installation:

Follow the remaining prompts to complete the installation. Once done, reboot your system.

Step 3: Log into the Shell

After rebooting, log into your system as the **root** user and its password you created during the installation. You'll be greeted with a shell prompt.

```
Debian GNU/Linux 12 debiannetinstall tty1
debiannetinstall login: root
Password:
Linux debiannetinstall 6.1.0-25-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.106-3 (2024-08-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@debiannetinstall:~# _
```

Login to Debian Minimal Install

Step 4: Create a Sudo User

During the installation, you will probably create a regular user. You should [assign sudo privileges](#) to that user now.

To do so, run:

```
apt install sudo
```

```
adduser ostechnix sudo
```

Replace ostechnix with your actual username.

Log out from the root session and log back in as the regular user.

From now on, you can perform administrative tasks as regular user with sudo privileges.

Step 5: Install GNOME Core

Now, let's install the core components of GNOME. Open the terminal and run the following commands:

```
sudo apt update
sudo apt install gnome-core -y
```

This will install the essential parts of GNOME without any extra bloat.

Step 6: Remove ifupdown Package

The Debian installer uses the `ifupdown` package for network management, but GNOME uses NetworkManager. To avoid conflicts and ensure your WiFi card works, remove `ifupdown`:

```
sudo apt purge ifupdown -y
```

Step 7: Reboot Your System

After removing `ifupdown`, reboot your system:

```
sudo shutdown -r now
```

Step 8: Configure NetworkManager

Once you're back in GNOME, you'll need to make a small change to the NetworkManager configuration file to ensure it manages your network devices correctly.

Edit the Configuration File:

```
sudo nano /etc/NetworkManager/NetworkManager.conf
```

Change the line `managed=false` to `managed=true`.

Press `Ctrl + O` to save the file, then `Ctrl + X` to exit the editor.

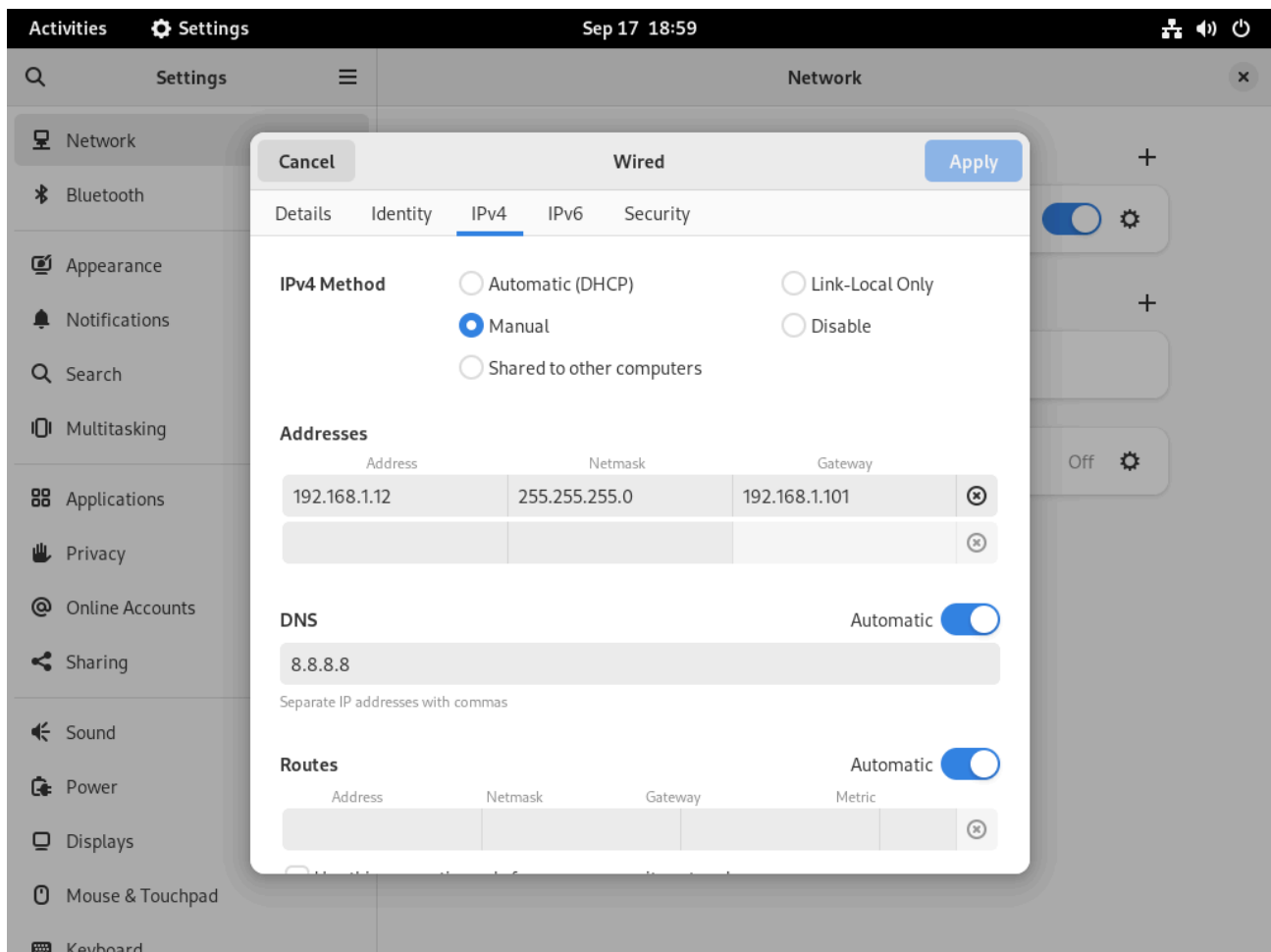
Reboot again:

```
sudo shutdown -r now
```

Step 9: Configure IP Address again

After log in to the system, you may need to **configure the IP address again**.

To do so, go to **Settings -> Network**. Click the gear button next to your network connection and configure the IP address for your network interface.



Configure IP Address in Debian

Step 10: Install Additional Tools (Optional)

At this point, you have a barebones GNOME environment. If you want to add more tools like a file manager, terminal, web browser, git, CLI downloaders, and text editor etc., you can do so by running:

```
sudo apt install gnome-session nautilus gnome-terminal firefox-esr git wget curl vim -y
```

This will install the GNOME session manager, the Nautilus file manager, GNOME Terminal, Mozilla Firefox, Git, Wget, Curl, and Vim editor.

Do not make your system bloat again by installing unnecessary software. Just install only the essential software you want in your Debian system.

Conclusion

Congratulations! You've successfully installed a minimal GNOME environment on Debian. This setup is lightweight and perfect for those who want a clean, efficient desktop experience. Feel free to customize it further by installing additional GNOME components or other software as needed. Enjoy your new Debian GNOME setup!

Related Read:

[How To Create Chroot Environments Using Mmdebstrap In Debian Linux](#)

GNOME Asia Summit 2024 will be Held in Bengaluru, India

Published: **Updated:** 319 views 3 mins read

The GNOME Foundation announced that **GNOME Asia Summit 2024** will be held in **Bengaluru, India**, from **December 6th to 8th**. This summit offers a fantastic opportunity for open-source enthusiasts, developers, and anyone interested in the GNOME project to come together. This year, GNOME Asia will have a **hybrid format**, allowing attendees to participate either **in person or virtually**.

What is GNOME Asia Summit?

GNOME Asia is an annual conference that centres around the GNOME project and the GNOME desktop environment. The summit serves as a platform for users, developers, foundation leaders, government representatives, and businesses to discuss current technologies and future developments within the GNOME ecosystem.

About GNOME

GNOME is a free and open-source software environment project backed by a non-profit foundation. The GNOME community and Foundation collaborate to create a user-friendly and efficient computing platform and software ecosystem built entirely on free software.

Why Bengaluru?

Bengaluru, often referred to as the "Silicon Valley of India," is a bustling city known for its thriving tech industry and active open-source community. The city is home to numerous Linux and GNOME contributors who play a crucial role in the global open-source movement.

But Bengaluru is more than just a tech hub; it's also a city with a rich cultural heritage. Attendees can explore landmarks like the Bangalore Palace, Tipu Sultan's Summer Palace, and the historic Bull Temple. For those seeking a moment of peace, the city offers serene green spaces such as Lalbagh Botanical Garden and Cubbon Park.

Call for Participation

The GNOME Foundation is currently welcoming talk and workshop proposals for GNOME Asia 2024. Both in-person and remote presentations will be considered. If you're interested in submitting a proposal, the deadline is **September 30th**.

The participants can give talks and conduct workshops on various topics related to GNOME and open-source software, including:

- Contributing to GNOME
- UI design
- Accessibility
- Human Interface Engineering (Icons and Graphic Design)
- Marketing/Engagement
- Developing GNOME on embedded systems or open hardware
- Contributing to Linux and FLOSS
- Linux kernel and development
- The development and promotion of open-source operating systems
- Distributions, including Debian, Fedora, openSUSE, Ubuntu, and FreeBSD
- The development and promotion of other open source projects
- Other topics related to Free/Libre and Open Source Software not listed above

Remember that GNOME Asia 2024 will be a hybrid event, so you can choose to present **in person or remotely**.

How to Apply

To give a talk or conduct a workshop at GNOME Asia 2024, you should submit a proposal through the event website by September 30th.

<https://events.gnome.org/event/258/abstracts/#submit-abstract>

Date and Venue

The GNOME Asia 2024 event will take place from **December 6th to 8th** in **Red Hat India Pvt. Ltd., Bengaluru**.

Date and Timing:

- **Start:** December 6, 2024, 9:00AM
- **End:** December 8, 2024, 4:00 PM

Venue Address:

Red Hat India Pvt. Ltd. Bagmane Constellation Business Park, Carina Building, 10th Floor East, Doddanekkundi, Bengaluru, Karnataka 560037.

Location Map: <https://maps.app.goo.gl/hDKFb9bczBh94N67A>